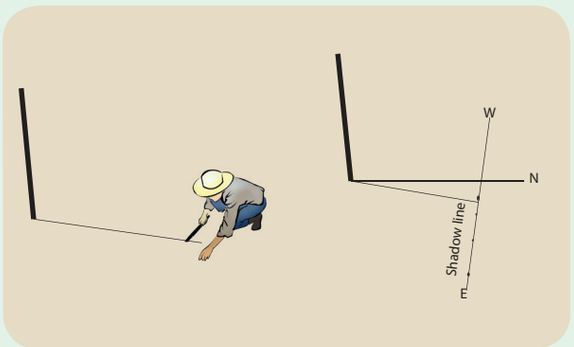
Fact File

Geo connects History

Secret to **Great Pyramid's** Near Perfect Alignment Possibly Found!

The Great Pyramid of Giza, 4,500 years ago, is an ancient feat of engineering. Now an archaeologist has figured out how the Egyptians may have aligned the pyramid almost perfectly along the cardinal points, north-south-east-west. Egyptians may have used **the autumn equinox**. Methods used by the ancient Egyptians to align the pyramids along the cardinal points are accurate.





On the day of the fall equinox, a surveyor placed a rod into the ground and tracked its shadow throughout the day. The result was a line running almost perfectly east-west. The Egyptians could have determined the day of the fall equinox by counting forward 91 days after the summer solstice.

umbra obscures the entire the moon's surface. Lunar eclipse can be seen from anywhere on the night side of the Earth. It lasts for a few hours due to the smaller size of the moon.

Phases of the Moon

The changing angles between the earth, the sun and the moon determine the phases of the moon. Phases of the moon (Figure 2.16) start from the **'New Moon'** every month. Then, only a part of the Moon is seen bright called **'Crescent'**, which develops into the **'first quarter'**. With the increasing brightness it turns into three quarters known as **'Gibbous'** and then it becomes a **'Full Moon'**. These stages are the waxing moon. After the **full moon**, the moon starts waning or receding through the stages of Gibbous, last quarter, crescent, and finally becomes invisible as dark New Moon.

The varying lengths of daylight in different latitudes

It is evident from the table that the duration of daylight is 12 hours throughout the year at the equator only. As one moves away from the equator, the seasonal variations in the duration of daylight increase. The seasonal variations in the duration of daylight are maximum at the polar region.

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